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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Cyber Security) (CBCS) Degree Examinations

May - June 2025

22CBS201 – FIREWALL AND UTM ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	What are Intrusion Detection Systems (IDS)? Demonstrate the types of IDS and their working'.	8	L*2	1	1
b)	Define Virtual Private Network (VPN)? Classify and explain the types of VPN with an example.	8	L2	1	1
c)	Explain the importance of network security.	4	L2	1	1
2. a)	Define network security. Outline the elements of network security.	8	L2	1	1
b)	Summarize the commonly used types of network security tools and software.	8	L2	1	1
c)	Explain common types of firewalls.	4	L2	1	1
	Unit – II				
3. a)	With a neat block diagram, explain the components of unified threat management architecture.	8	L2	2	1
b)	Compare Stateless Packet Filtering Firewalls versus Stateful Packet Filtering Firewalls.	6	L2	2	1
c)	Outline the architecture layers of Unified Threat Management (UTM).	6	L2	2	1
4. a)	Explain Intrusion Prevention System (IPS) and its key concepts.	8	L2	2	1
b)	Define content filtering and Outline key types of content filtering.	6	L2	2	1
c)	What is Web Application Firewall (WAF)? Explain key features and functions of a WAF.	6	L2	2	1
	Unit – III				
5. a)	Classify the intrusion prevention systems and explain their working and benefits provided by them.	10	L2	3	1
b)	Define URL filtering. Summarize types of URL filtering.	10	L2	3	1
6. a)	Outline perimeter-based security and zero trust security architecture along with their components.	10	L2	3	1
b)	Justify how UTM architecture is better than other architectures?	10	L2	3	1,2
	Unit – IV				
7. a)	Highlight the importance of monitoring firewall logs. Outline step-by-step guide on how you can monitor firewall logs effectively.	10	L2	4	1
b)	Discuss how to Test firewall security.	10	L2	4	1
8. a)	Define document firewall configuration. Summarize the steps to document your firewall configuration effectively.	10	L2	4	1
b)	Outline the generic troubleshooting steps.	10	L2	4	1

P.T.O

Unit – V

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| 9. | a) | Explain Anti-Bot Technology in NGFWs. | 10 | L2 | 5 | 1 |
| | b) | Define threat analysis and assessment. Outline the threat analysis and the threat assessment process. | 10 | L2 | 5 | 1 |
| 10. | a) | How threat sharing works? Explain the key benefits of threat sharing and collaboration. | 10 | L2 | 5 | 1 |
| | b) | Define real-time threat response. Explain the key components of real-time threat response. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
